SMART MANHOLE UTILITIES URBAN IOT

Autonomous Sub-Surface Infrastructure Monitoring, Multi-Sensor Telemetry & 3D Digital Twin SCADA Platform

PROJECT GROUP: P077	2400040006 P V Kishor Kumar Reddy Lead Firmware & SCADA	2400040004 T Perumal Sensors & Conditioning	2400040303 Chundu Uday Sai IoT Gateway & Cloud	2400040339 Ayan Chowdhury Hardware & Power Testing	LIVE SCADA TWIN pvkkr.github.io/smart-manhole-twin/

1. OBJECTIVES & URBAN APPS

The Urban Sub-Surface Crisis: Unmonitored utility manholes cause severe urban flooding from hidden blockages, lethal toxic gas asphyxiation (H■S, Methane) for sanitation workers, and catastrophic vehicular accidents from displaced, tilted, or stolen covers.

Key Project Objectives:

- **Continuous Sub-surface Telemetry:** Autonomous 24/7 sensing of water depth, gas concentrations, temperature, and cover orientation.
- **Dual-Mode Latency Pipeline:** Sub-20ms local SCADA WebSocket telemetry + 15s historical cloud sync to ThingSpeak.
- **3D Interactive Digital Twin:** Physics-accurate visual twin rendering real-time lid kinematics and water surcharge.
- **Immediate Edge Alarms:** Onboard acoustic alarm and physical 16x2 LCD display alerting pedestrians on-site.

Real-Time Urban Applications:

- **Municipal Flood Management:** Early flash-flood alerts before storm surge overflows onto city streets.
- **Sanitation Worker Safety:** Pre-entry gas hazard screening preventing fatal sewer asphyxiation.
- **Anti-Theft & Critical Infrastructure:** Instant tamper alarms guarding fiber-optic and power conduits.
- **Smart City Traffic Safety:** Real-time road hazard dispatch when vehicles dislodge heavy cast-iron lids.

2. HARDWARE ARCHITECTURE

Dual-Core Distributed Processing:

- **Edge Master: STM32F103C8T6 (ARM Cortex-M3 @ 72MHz):** Executes bare-metal register-level CMSIS firmware. Cycles a 10Hz non-blocking deterministic sensor loop with zero RTOS overhead.
- **IoT Gateway: NodeMCU V3 (ESP8266):** Ingests JSON frames over USART1 (PA9 → D7 @ 115200 baud). Hosts local REST/WebSocket server and posts to ThingSpeak Cloud.

Sensor Cluster & Pin Interconnect:

Sensor	STM32 Pin	Function / Range
HC-SR04	PA1 (T), PA2 (E)	Water Air Gap (2-400cm)
MQ-2 Gas	PA0 (ADC1_IN0)	Toxic Gas (0-4095 ADC)
MPU-6050	PB8 (SCL), PB9 (SDA)	3-Axis Tilt (0° to 90°)
16x2 LCD	PB8 (SCL), PB9 (SDA)	All-in-One Live Readout
DS18B20	PB12 (1-Wire)	Sub-surface Temp (°C)
Reed Switch	PB13 (Pull-Up)	Cover Seal / Tamper
Buzzer	PB0 (GPIO Out)	Audible Warning Beacon

- **Echo Voltage Divider:** 1kΩ / 2kΩ resistive divider safely steps 5.0V Echo down to 3.33V for ADC pin PA2 protection.

3. BOTTLENECKS SOLVED

Key Engineering Challenges & Fixes:

- **GCC nano.specs Float Omission Bug:** Default ARM GCC nano.specs stripped %f support in snprintf, causing blank LCD numbers (D:cm, T:C) and broken JSON. *Fix: Implemented fixed-point integer formatting (%d.%d).*
- **Bit-Bang I2C Repeated-Start Glitch:** MPU6050 ACK held SDA low during repeated starts, causing read timeouts and locking tilt to 0°. *Fix: Replaced with clean Stop-Start cycle with settle delays.*
- **DS18B20 750ms Blocking Freeze:** 12-bit temp conversion froze MCU execution for 750ms per loop. *Fix: Re-engineered as an asynchronous non-blocking state machine cycling at 10Hz.*
- **Browser / CDN 300-Second Stale Cache:** Browsers cached ThingSpeak feeds/last.json for 5 minutes. *Fix: Appended dynamic timestamp query params (?_=${Date.now()}) and strict no-store headers.*
- **HC-SR04 Air Gap Clamping:** Raw distance was artificially clamped <=50cm. *Fix: Expanded to full 400cm range with ARM Cortex DWT cycle-counter microsecond timing.*

4. SCADA TWIN & FUTURE WORK

3D Digital Twin Features:

- **Three.js Kinematics:** 3D manhole model mirrors physical lid tilt angle in real time based on MPU6050 readings.
- **Dynamic Water Physics:** Shader-rendered fluid volume calculates water percentage from HC-SR04 depth.
- **Multi-Channel SCADA Alarms:** Visual color-coded cards and audible sirens trigger on CRITICAL thresholds.

Physical 16x2 LCD Interface:

Simultaneous 5-parameter real-time display:

```
Row 1: D: 25cm G:790
Row 2: T:28C TL: 0 SEAL (or TL:35 OPEN)
```

Future Improvements & Scaling:

- **LoRaWAN / NB-IoT Long Range:** Subterranean transmission through heavy cast-iron lids to gateways 10km away with 5+ year battery lifespan.
- **Energy Harvesting:** Piezoelectric generators harvesting kinetic vibration from traffic rolling over covers.
- **Edge TinyML Anomaly Detection:** Microcontroller-level predictive analytics forecasting sewer pipe siltation.
- **City GIS Integration:** Spatial fleet-wide mapping connecting thousands of manholes to municipal command centers.